

1 Power series

Question 1. Determine the interval of convergence of the following power series.

1. $\sum_{n=0}^{\infty} \frac{x^n}{3^n}$ By geometric series, it converges precisely when $-1 < \frac{x}{3} < 1 \Leftrightarrow -3 < x < 3$.
 $\left(\frac{x}{3}\right)^n$ $\boxed{(-3, 3)}$

2. $\sum_{n=0}^{\infty} \frac{(x-1)^n}{3^n}$ Shifted version of part 1.
 $-1 < \frac{x-1}{3} < 1 \Leftrightarrow -2 < x < 4$
 $\boxed{(-2, 4)}$

3. $\sum_{n=0}^{\infty} \frac{x^{2n}}{3^n}$ By geometric series, it converges precisely when $-1 < \frac{x^2}{3} < 1$
 $\Leftrightarrow \frac{x^2}{3} < 1$ $\boxed{(-\sqrt{3}, \sqrt{3})}$
 $\Leftrightarrow -\sqrt{3} < x < \sqrt{3}$

4. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ By ratio test, we need $\lim_{n \rightarrow \infty} \frac{x^{2n+2}}{(2n+2)!} \cdot \frac{(2n)!}{x^{2n}} < 1$
 but $\lim_{n \rightarrow \infty} \frac{x^2}{(2n+2)(2n+1)} = 0 < 1$ for all x .
 So $\boxed{(-\infty, \infty)}$

Note: this is $\arctan(x)$
 5. $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$ First, check for absolute convergence. By ratio test, we need $\lim_{n \rightarrow \infty} \frac{|x|^{2n+3}}{2n+3} \cdot \frac{2n+1}{|x|^{2n+1}} < 1$
 $\lim_{n \rightarrow \infty} \frac{2n+1}{2n+3} |x|^2 = |x|^2$. So it converges absolutely on $(-1, 1)$.
 Then, check end points, we see by AST, the series converges on $\boxed{[-1, 1]}$

6. $\sum_{n=1}^{\infty} \frac{x^n}{2 \cdot 4 \cdot 6 \cdots (2n)}$ By ratio test, we need $\lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{2 \cdot 4 \cdots (2n+2)} \cdot \frac{2 \cdot 4 \cdots (2n)}{|x|^n} < 1$
 $\lim_{n \rightarrow \infty} \frac{|x|}{2n+2} = 0$ for all x . So $\boxed{(-\infty, \infty)}$
 Note: recognize that it is $\sum_{n=1}^{\infty} \frac{x^n}{n! 2^n} = e^{x/2} - 1$

7. $\sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n} x^n$ First check for absolute convergence. By ratio test, we need $\lim_{n \rightarrow \infty} \frac{2|x|}{n+1} \cdot n = 2|x| < 1$.
 $\Rightarrow |x| < \frac{1}{2}$. So it converges absolutely on $-\frac{1}{2} < x < \frac{1}{2}$.
 Check endpoints: $x = -\frac{1}{2} \Rightarrow$ diverges by p-test
 $x = \frac{1}{2} \Rightarrow$ converges by AST
 $\boxed{(-\frac{1}{2}, \frac{1}{2})}$

8. $\sum_{n=0}^{\infty} \frac{x^n}{\sqrt{2n+1}}$ First, check for absolute convergence. By ratio test, we need $\lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{\sqrt{2n+3}} \cdot \frac{\sqrt{2n+1}}{|x|^n} = |x| < 1$. Thus, it converges absolutely for $-1 < x < 1$.
 At endpoints: $x = 1 \Rightarrow$ diverges by comparison with p-series
 $x = -1 \Rightarrow$ converges by AST.
 So $\boxed{(-1, 1)}$

2 Maclaurin/Taylor series

Question 2. Find the Maclaurin series (Taylor series centered at $x = 0$) for the following functions.

1. $f(x) = \frac{1}{3+x} = \frac{1}{3} \left(\frac{1}{1+x/3} \right) = \frac{1}{3} \sum_{n=0}^{\infty} \left(-\frac{x}{3} \right)^n$
 $= \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} x^n$

2. $f(x) = e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}$

3. $f(x) = (x+1)^3 + (x+1)^2 + (x+1) + 1$
 expanding = $x^3 + 3x^2 + 3x + 1 + x^2 + 2x + 1 + x + 1 + 1$
 $= x^3 + 4x^2 + 4x + 4$

4. $f(x) = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^n + (-x)^n}{2(n!)} = \sum_{n=0}^{\infty} \frac{(1+(-1)^n) x^n}{2(n!)}$
 Note: this is $\cosh(x)$, same as 1.4.
 $= \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$ ($n=2k$)

5. $f(x) = \frac{d^2}{dx^2} (x \cos(x^2)) = \frac{d^2}{dx^2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+1}}{(2n)!}$
 $= \sum_{n=0}^{\infty} \frac{(-1)^n (4n+1)(4n) x^{4n-1}}{(2n)!}$

6. $f(x) = \int_0^x \frac{\sin t}{t} dt = \int_0^x \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n+1)!} dt$
 $= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1) \cdot (2n+1)!}$

7. (challenge) $f(x) = \int_{-2}^x \frac{1}{t^2+4} dt$
 $= \frac{1}{4} \int_{-2}^x \frac{1}{1+(t/2)^2} dt$ $t/2 = u$
 $= \frac{1}{2} \int_{-1}^{x/2} \frac{1}{1+u^2} du$ $du = 2dt$
 $= \frac{1}{2} \arctan(u) \Big|_{-1}^{x/2} = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + \frac{\pi}{8}$
 $= \frac{\pi}{8} + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1) 2^{2n+2}}$